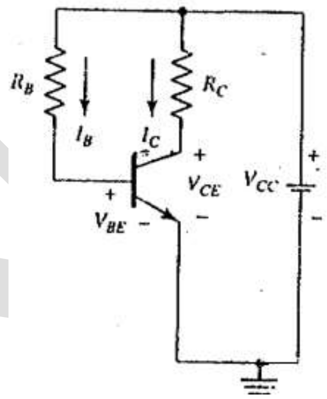
Module-1(a) - Transistor Biasing

Syllabus - Transistor Biasing: Voltage Divider Bias, VDB Analysis, VDB Load line and Q point, Two supply Emitter Bias, Other types of Bias.

Introduction

DC load line : - The straight line on the output characteristics of a transistor which gives the values of output current I_C and output voltage V_{CE} corresponding to zero signal input is called the DC load line.

Consider an n-p-n transistor biased as shown in such that it is in active region.



Applying KVL to output circuit, $V_{CC} - I_C R_C - V_{CE} = 0$

$$I_C = \left(-\frac{1}{R_C}\right) V_{CE} + \frac{V_{CC}}{R_C} \quad \text{------------ (1)}$$

Equation (1) represents a straight line of the form y = mx + C with slope m= $\left(-\frac{1}{R_C}\right)$ and constant C = $\left(\frac{V_{CC}}{R_C}\right)$

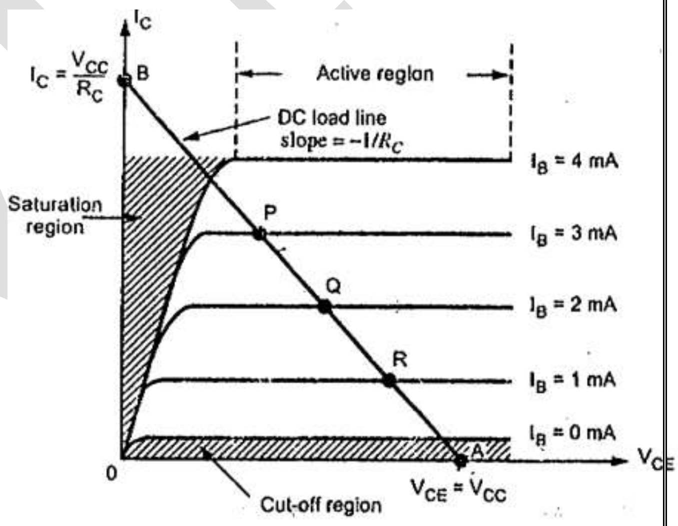
When $V_{CE} = V_{CC}$ ➔ $I_C = 0$ --- Point A on the V_{CE} axis (X- axis)

When $V_{CE} = 0$, ➔ $I_C = \frac{V_{CC}}{R_C}$ ----- Point B on the I_C axis (Y axis)

The line AB is the DC load line with slope $\left(-\frac{1}{R_C}\right)$

DC bias point or Quiescent Point (Q-point)

The intersection of DC load line and output characteristics of a transistor is called Q point or Operating point for a particular value of the input base current I_B.



P, Q & R are Quiescent points for currents I_{B3}, I_{B2} & I_{B1} respectively.

When an input signal is applied to the base of the transistor I_B varies according to the instantaneous amplitude of the signal. This causes I_C to vary and consequently produces a variation in V_{CE}. The Q point should be selected midway between the saturation and cut-off regions be placed either in active, cut-off or in saturation regions.

Transistor Biasing:

Transistor biasing means applying suitable external dc voltages to transistor terminals to establish a constant dc current in the collector of the transistor so that this current is calculable, predictable and insensitive to variation in temperature and other transistor parameters.

Transistor biasing is required to maintain the operating point in the middle of the active region so that maximum output signal swing can take place and Q point will not shift into saturation or cut-off region.

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Requirements of Biasing Circuits

(1) E - B junction should be forward biased and C – B junction should be reverse biased i,e the transistor should be operated in the active region. (2) The circuit should provide temperature stability. (3) The operating point should be made independent of transistor parameters α, β and γ.

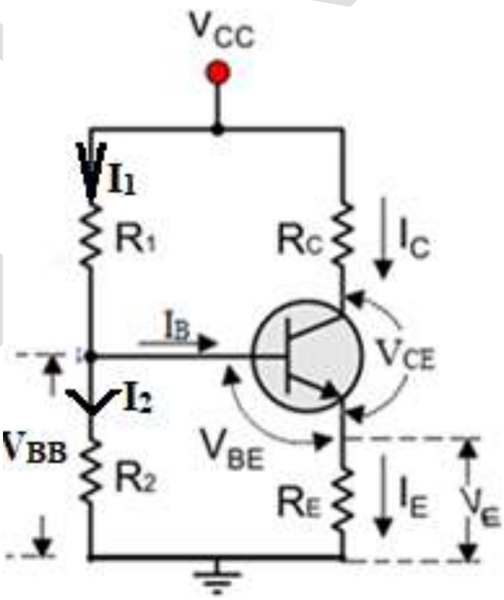
There are mainly three types of biasing circuits – Fixed bias or Base bias circuit, Collector to Base bias circuit and Voltage Divider Bias (VDB) circuit.

Voltage Divider Bias (VDB) Circuit

Introduction: Voltage divider bias is also known as Emitter current bias which gives the most stable operating point when compared to base bias and collector to base bias schemes. In this scheme, the levels of I_C and V_{CE} are almost independent of β of the transistor.

In this scheme, a single power supply and a voltage divider network consisting of two resistors R_1 and R_2 are used for supplying a fraction of the supply voltage V_{CC} to bias the transistor. The additional resistor R_E provides the stability.

Simplified or Approximate Analysis



In approximate analysis, the base current is assumed to be much smaller than the voltage divider current I_2. The voltage divider network consisting of two resistors R_1 and R_2 are used to produce the base voltage V_B from the supply voltage V_{CC} such that $I_2 >>> I_B$ so that $I_1 = I_2 + I_B \cong I_2$ ----------- (1)

Applying KVL for the path ($V_{CC} \rightarrow R_1 \rightarrow R_2$)

$V_{CC} - I_1R_1 - I_2R_2 = 0$ and from equation (1) using $I_1 \cong I_2$

$I_2(R_1 + R_2) = V_{CC}$ $\rightarrow$ $I_2 = \left(\frac{V_{CC}}{R_1+R_2}\right)$

$V_{BB} = V_{R2} = I_2 R_2 = \left(\frac{V_{CC}\ R_2}{R_1+R_2}\right)$ and voltage across R_E is given by

$V_E = I_ER_E$

Applying KVL to base-emitter loop,

$V_{BB} - V_{BE} - V_E = 0$ or $V_E = I_E R_E = V_{BB} - V_{BE}$

$$I_E = \left(\frac{V_{BB} - V_{BE}}{R_E}\right) = I_C$$

Applying KVL to the collector-emitter loop,

$\rightarrow$ $V_{CC} - I_CR_C - V_{CE} - I_ER_E = 0$

Using $I_E \cong I_C$; solving $V_{CE} = V_{CC} - I_C(R_C + R_E)$ or $I_C = \left(\frac{V_{CC} - V_{CE}}{R_C+R_E}\right)$

The Q point is located at (V_{CE}, I_C)

The calculations in this analysis do not depend on changes in the transistor, the collector current, or the temperature. The *Q* point of this circuit is stable

Problem : Determine the collector to emitter voltage for the VDB circuit with R_1 = 10kΩ; R_2 = 2.2kΩ; R_C = 3.3kΩ; R_E = 1kΩ and V_{CC} = 10V using simplified analysis.

$V_{BB} = \left(\frac{V_{CC}\ R_2}{R_1+R_2}\right) = \left(\frac{10(2.2)}{10+2.2}\right) = 1.8V$

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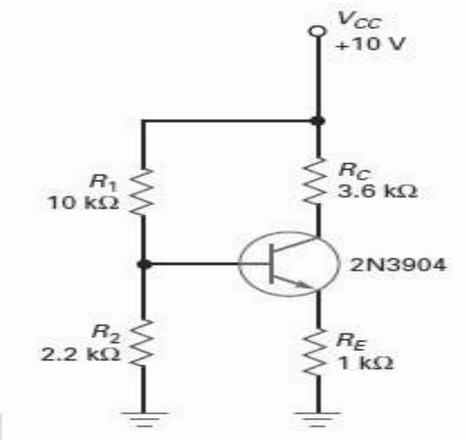


Voltage across R_E is given by

$V_E = V_{BB} - V_{BE} = 1.8 - 0.7 = 1.1V$

$$I_E = \left(\frac{V_E}{R_E}\right) = \left(\frac{1.1}{1}\right) = 1.1mA = I_C$$

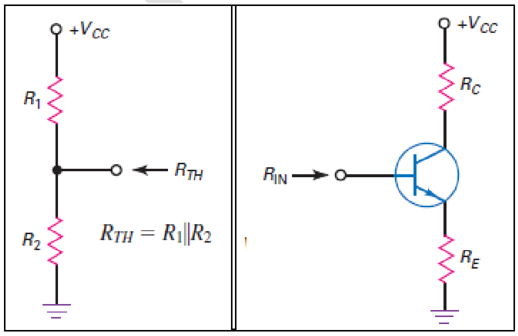
Applying KVL to the collector-emitter loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

Using $I_E \cong I_C$; solving $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$V_{CE} = 10 - 1.1(3.6 + 1) = 4.94V$

The Q point is located at (4.94V, 1.1mA)



Stiff Voltage Source

In a Stiff Voltage Source, the source resistance R_S is at least 100 times smaller than the load resistance R_L.

$R_S < 0.01R_L$ ------ 100:1 rule

Under this condition, the source resistance R_S can be ignored

Stiff Voltage Divider

In a Stiff Voltage Divider, the source resistance R_S is at least 100 times smaller than the load resistance R_L translates into

$(R_1 \| R_2) < 0.01R_{in}$ where $R_{in} = \beta_{dc} R_E$

Firm Voltage Divider

In a firm Voltage Divider, the source resistance R_S is at least 10 times smaller than the load resistance R_L (10: 1) translates into $(R_1 \| R_2) < 0.1R_{in}$ where $R_{in} = \beta_{dc} R_E$

Accurate VDB Analysis

Figure shows a VDB circuit. In this method, the base current I_B is considered. The power supply V_{CC}, the voltage divider network ($R_1 - R_2$) are replaced with a Thevenin's equivalent circuit consisting of a voltage source V_{TH} in series with a resistance R_{TH} between V_{CC} and ground.

$$V_{BB} = \left[\frac{V_{CC}\,R_2}{R_1 + R_2}\right] \quad \text{and} \quad R_B = R_{TH} = R_1||R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

The modified circuit is shown with voltage source V_{BB} and base resistor R_B.

Applying KVL to the base circuit

$$V_{BB} - R_B I_B - V_{BE} - I_E R_E = 0 \quad \text{or} \quad V_{BB} - R_B\left(\frac{I_E}{\beta+1}\right) - V_{BE} - I_E R_E = 0$$

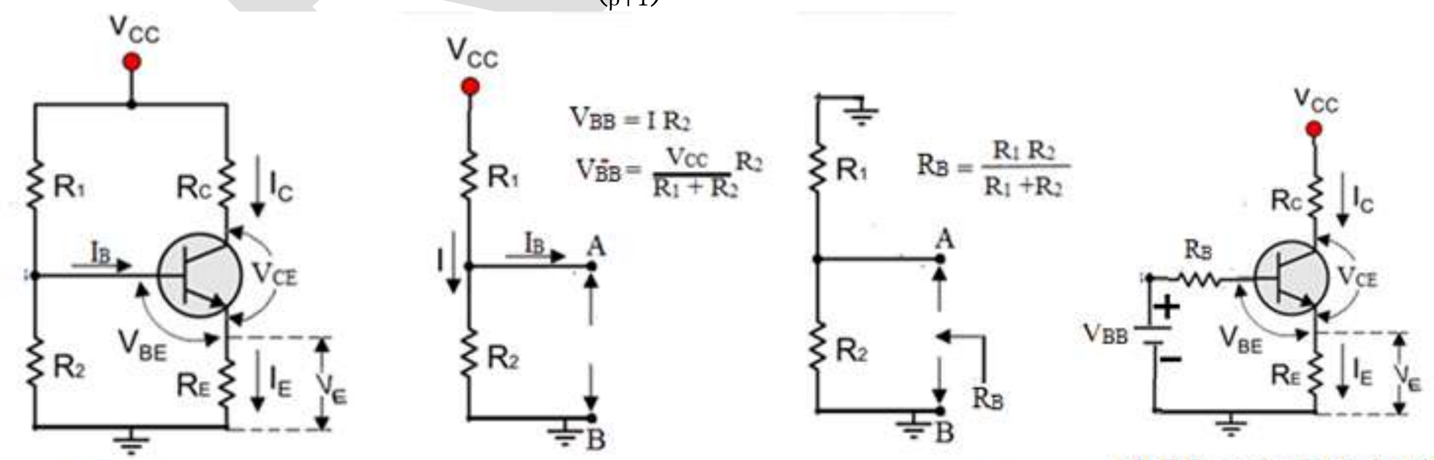


Fig: (a) Voltage divider bias circuit (b) Calculation of V_{BB} (c) Calculation of R_B **(d) DC equivalent circuit**

Emitter Current $I_E = \left[\dfrac{V_{BB} - V_{BE}}{R_E + \left(\dfrac{R_B}{\beta + 1}\right)}\right]$ and Base current $I_B = \left(\dfrac{I_E}{\beta + 1}\right) = \left[\dfrac{V_{BB} - V_{BE}}{R_B + (1+\beta)R_E}\right]$

$I_C = I_E - I_B$

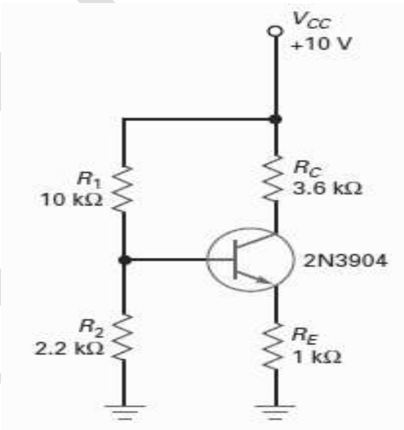
Applying KVL to collector –emitter loop, $V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$

Solving for V_{CE} , $V_{CE} = V_{CC} - I_C R_C + I_E R_E)$

➔ The resistor R_E provides the stability the current I_E by negative feedback action shown below.

$$I_E\uparrow \longrightarrow I_E R_E \uparrow \longrightarrow V_E \uparrow \longrightarrow V_{BB} \longrightarrow V_{BE}\downarrow \longrightarrow I_C \downarrow \longrightarrow I_E \downarrow$$

Problem – Is the Voltage divider in the circuit shown is stiff? Calculate the accurate value of the emitter current. Given β_{dc} = 200.



Solution - $R_{in} = \beta_{dc} R_E = 200(1) = 200\ k\Omega$ and $0.01\ R_{in} = 2\ k\Omega$

$R_{Th} = [R_1 \parallel R_2] = (10 \parallel 2.2) = 1.8\ k\Omega$ and $V_{BB} = \left[\dfrac{V_{CC}\ R_2}{R_1 + R_2}\right] = \left[\dfrac{10(2.2}{10+2.2}\right] = 1.8V$

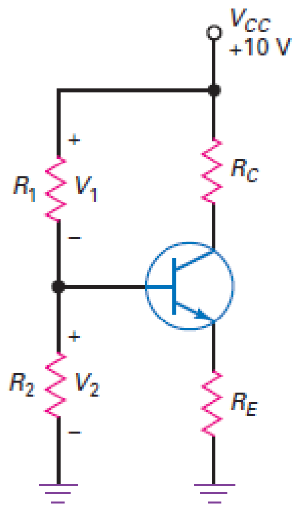
Since 100:1 rule $(R_1 \parallel R_2) < 0.01R_{in}$ is satisfied, the **Voltage divider in the circuit is stiff.**

$$I_E = \left[\frac{V_{BB} - V_{BE}}{R_E + \left(\frac{R_B}{\beta + 1}\right)}\right] = \left[\frac{1.8 - 0.7}{1 + \left(\frac{1.8}{200 + 1}\right)}\right] = 1.09\ mA$$

VDB Design Guidelines

Choose $V_E = 0.1V_{CC}$	**Choose $V_{CE} = 0.5V_{CC}$**	**Using 100:1 rule**	$V_2 = 0.7 + V_E$
$R_E = \left(\frac{V_E}{I_E}\right)$ and $I_E \cong I_C$	**$R_C = 4\ R_E$**	$R_{TH} \leq 0.01\beta_{dc} R_E$ $R_2 \leq 0.1\beta_{dc} R_E$	$V_1 = V_{CC} - V_2$ $R_1 = \left(\frac{V_1}{V_2}\right) R_2$

Problem - Design the voltage divider bias circuit to meet the following specifications V_{CC} =10V, V_{CE} at mid point, stiff voltage divider, collector current 1mA, β_{dc} = 70 – 200



For stiff voltage divider bias circuit,

V_E = 10% of $V_{CC} = 0.1(V_{CC}) = 0.1(10) = 1V$ and $I_E \cong I_C = 1mA$

$R_E = \left(\frac{V_E}{I_E}\right) = \left(\frac{1}{1}\right) = 1\ k\Omega$ $\quad R_C = 4\ R_E = 4(1) = 4\ k\Omega$

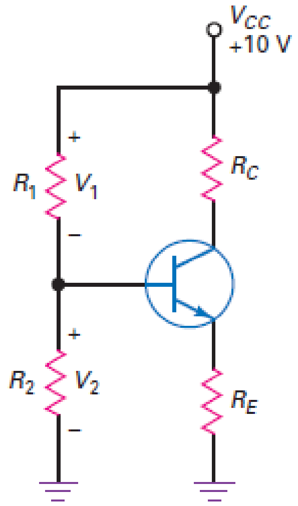
$R_2 \leq 0.01\ R_E \beta_{dc(lower)}$ $\quad R_2 = 0.01(70)(1) = 700\ \Omega$

$V_2 = 0.7 + V_E = 0.7 + 1 = 1.7V$

$V_1 = V_{CC} - V_2 = 10 - 1.7 = 8.3V$

$R_1 = \left(\frac{V_1}{V_2}\right) R_2 = \left(\frac{8.3}{1.7}\right)(700) = 3.41\ k\Omega$

Problem - Design the voltage divider bias circuit to meet the following specifications V_{CC} =10V, V_{CE} at mid point, stiff voltage divider, collector current 10mA, β_{dc} = 100 – 300



For stiff voltage divider bias circuit,

V_E = 10% of V_{CC} = $0.1(V_{CC}) = 0.1(10) = 1V$ and $I_E \cong I_C = 10mA$

$R_E = \left(\frac{V_E}{I_E}\right) = \left(\frac{1}{10}\right) = 100\ \Omega$ $\quad R_C = 4\ R_E = 4(100) = 400\ \Omega$

$R_2 \leq 0.01\beta_{dc(lower)}R_E$ $\quad R_2 = 0.01(100)(100) = 100\ \Omega$

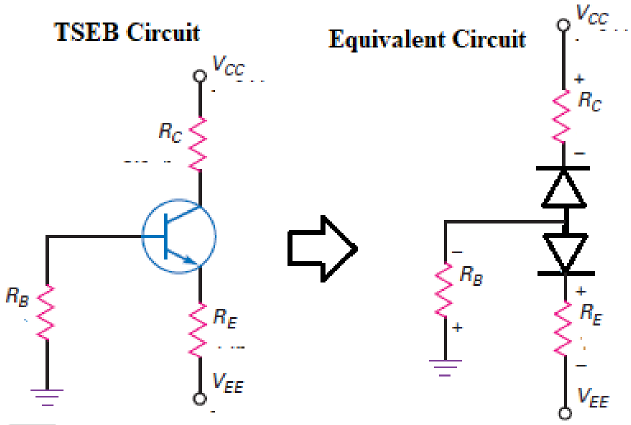
$V_2 = 0.7+V_E = 0.7+1 = 1.7V$

$V_1 = V_{CC} - V_2 = 10 - 1.7 = 8.3V$

$R_1 = \left(\frac{V_1}{V_2}\right) R_2 = \left(\frac{8.3}{1.7}\right)(100) = 488\ \Omega$

Two supply Emitter Bias (TSEB) Circuit

The circuit uses two power supplies, the positive supply V_{CC} reverse biases the collector diode and negative supply V_{EE} forward biases the emitter diode.



Applying KVL to emitter – base loop

$V_{EE} + V_E + V_{BE} = 0$ or $V_E = -V_{BE} - V_{EE}$ and

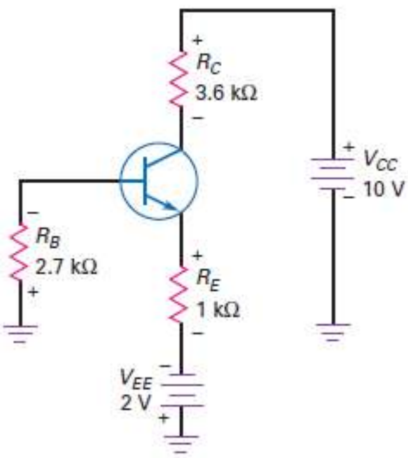
$I_E = \left(\frac{V_E}{R_E}\right) = \left(\frac{-V_{BE} - V_{EE}}{R_E}\right)$

The collector voltage $V_C = V_{CC} - I_C R_C$

and $V_{CE} = V_C + V_{BE}$ Choose $R_B \leq 0.01\beta_{dc}R_E$

Since R_B is grounded, voltage across it $V_B \cong 0$

Problem – For the TSEB circuit shown, determine the collector voltage. If the resistance R_E is increased to 1.8 kΩ, find the new value of the collector voltage.



Solution <u>For R_E = 1kΩ</u>

$I_E = \left(\frac{V_E}{R_E}\right) = \left(\frac{-V_{BE} - V_{EE}}{R_E}\right) = \left(\frac{-0.7-(-2)}{1}\right) = 1.3\ mA = I_C$

The collector voltage $V_C = V_{CC} - I_C R_C = 10 - 1.3(3.6) = 5.32\ V$

and $V_{CE} = V_C + V_{BE} = 5.32 + 0.7 = 6.02V$

<u>For R_E = 1.8 kΩ</u>

$I_E = \left(\frac{V_E}{R_E}\right) = \left(\frac{-V_{BE} - V_{EE}}{R_E}\right) = \left(\frac{-0.7-(-2)}{1.8}\right) = 0.722\ mA = I_C$

The collector voltage $V_C = V_{CC} - I_C R_C = 10 - 0.722(3.6) = 7.4\ V$

and $V_{CE} = V_C + V_{BE} = 7.4 + 0.7 = 8.1\ V$

Other types of Biasing Circuits

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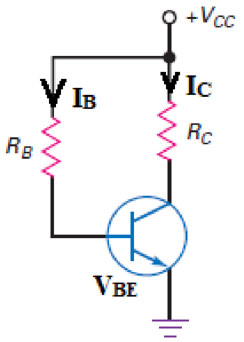
(a) Base Bias or Fixed bias



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<u>DC Analysis</u>

Applying KVL to Base loop,

$V_{CC} - I_B R_B - V_{BE} = 0$

$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ and $I_C = \beta I_B$

KVL to Collector emitter loop,

$V_C = V_{CE} = V_{CC} - I_C R_C$

The Q point is located at (V_{CEQ}, I_{CQ})

Q point is sensitive to changes in the value of β

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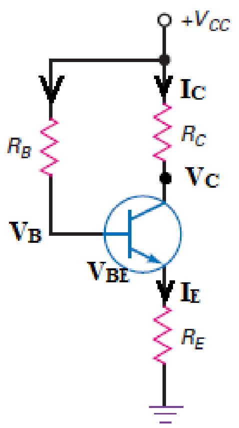
(b) Emitter Feedback Bias



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DC analysis

Applying KVL to Base loop,

$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$ and using $I_B \cong (I_E/\beta)$

$I_B = \frac{V_{CC} - V_{BE}}{R_E + \left(\frac{R_B}{\beta}\right)}$ $V_E = I_B R_E$ and $V_B = V_{BB} + V_E$

KVL to Collector emitter loop,

$V_C = V_{CC} - I_C R_C$ $V_{CE} = V_{CC} - I_C R_C - V_E$

The Q point is located at (V_{CEQ}, I_{CQ})

The change in the collector voltage is feedback to the base current and this feedback is called negative feedback as it opposes the change in the collector current. Q point is sensitive to changes in the value of β

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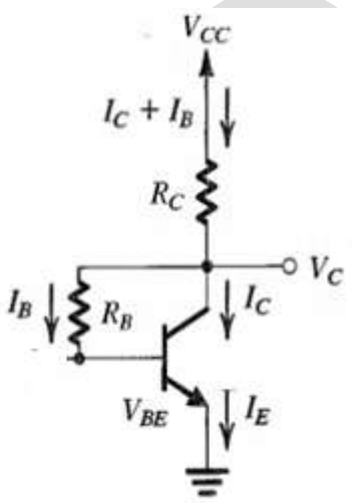
(c) Collector Feedback Bias



Also called as Self bias.

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<u>DC Analysis</u>

Applying KVL to the Collector – base loop,

$V_{CC} - (I_C + I_B) R_C - I_B R_B - V_{BE} = 0$

$V_{CC} - (\beta I_B + I_B) R_C - I_B R_B - V_{BE} = 0$

$I_B = \left[\frac{V_{CC} - V_{BE}}{(1+\beta)R_C + R_B}\right]$ and Emitter Current $I_E \cong \beta I_B = \left[\frac{V_{CC} - V_{BE}}{R_C + \frac{R_B}{\beta}}\right]$

Also applying KVL to collector to emitter via R_B

$-I_B R_B - V_{BE} + V_{CE} = 0$ or $I_B = \left[\frac{V_{CE} - V_{BE}}{R_B}\right]$

Applying KVL to Collector – Emitter loop

$V_{CC} - (I_C + I_B) R_C - V_{CE} = 0$ ➔ $V_{CE} = V_{CC} - (I_C + I_B) R_C$

$(I_C + I_B) = I_E = \frac{V_{CC} - V_{CE}}{R_C}$ ➔ $I_C = (I_E - I_B)$

Also $V_{CB} = I_B R_B = \left[\frac{I_E R_B}{\beta}\right]$

The resistor R_B provides the stability the current I_E by negative feedback action A small feedback voltage is applied to the base in order to neutralize any change in the collector current.

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Review Questions

1. What is biasing a transistor. Mention the requirements for biasing a transistor.
2. Define (i) DC load line (ii) Q point. Explain how dc load line is drawn and q point is located on the dc load-line.
3. Explain voltage divider bias circuit using (i) Approximate analysis and (ii) Accurate analysis
4. Write an explanatory note on (i) Base bias (ii) Emitter feedback bias (iii) Collector to base bias or collector feedback bias.